PLASMA LIPIDS AND PLASMA LIPOPROTEINS

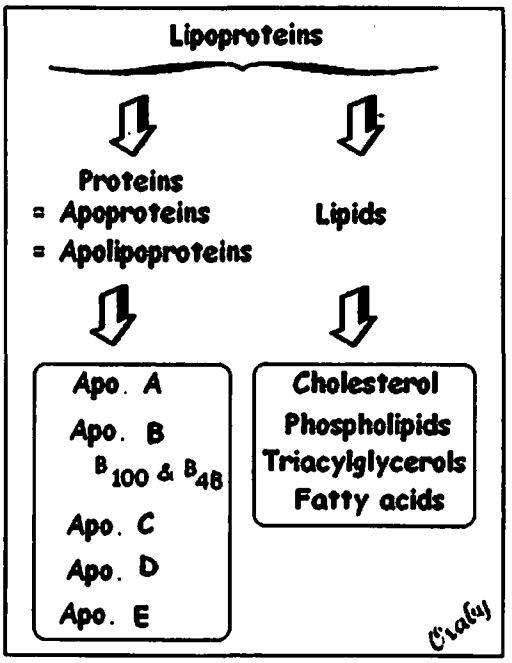
I. *PLASMA LIPIDS:* During fasting, total plasma lipids concentration ranges from 360-820 mg/dl. They include:

1-Cholesterol :	**140-220 mg/dl** **70% Cholesteryl esters** **30% Free cholesterol**
2-Phospholipids :	150-200 mg/dl
3-Triacylglycerols :	40-160 mg/dl
4-Free fatty acids :	6-16 mg/dl

II. *PLASMA LIPOPROTEINS :*

A. <u>Importance:</u>

1. Lipids alone are water insoluble compounds. Thus, they cannot be transported in plasma.
2. Lipids are conjugated to proteins to form lipoproteins, which are water-soluble and can be transported in plasma.
3. These proteins are synthesized by the liver and called apolipoproteins (or apoproteins). They are five classes: A, B, C, D and E.
4. Failure of liver to synthesize apolipoproteins leads to accumulation of fat in liver and this condition is called fatty liver (see later).



B. Methods of separation of plasma lipoproteins:

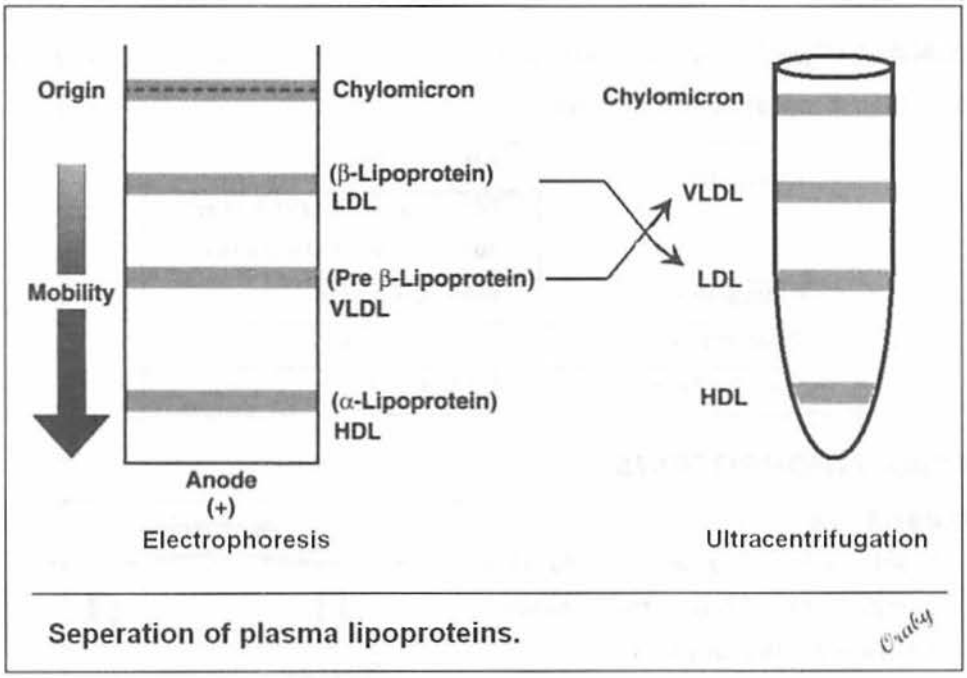
1. Ultracentrifugation:

a) Ultracentrifugation means centrifugation of compounds at high speed ~ 40,000 rounds per minutes (RPM).

b) By ultracentrifugation, plasma lipoproteins are separated into five fractions according to their density. These are chylomicrons, very low density lipoprotein (VLDL), low density lipoprotein (LDL), high density lipoprotein (HDL) and albumin free fatty acids (FFA) complex.

2. Electrophoresis :

a) By electrophoresis, plasma lipoproteins are separated into five fractions. These are chylomicrons, β-lipoprotein, pre β-lipoprotein, α-lipoprotein and albumin-FFA complex.



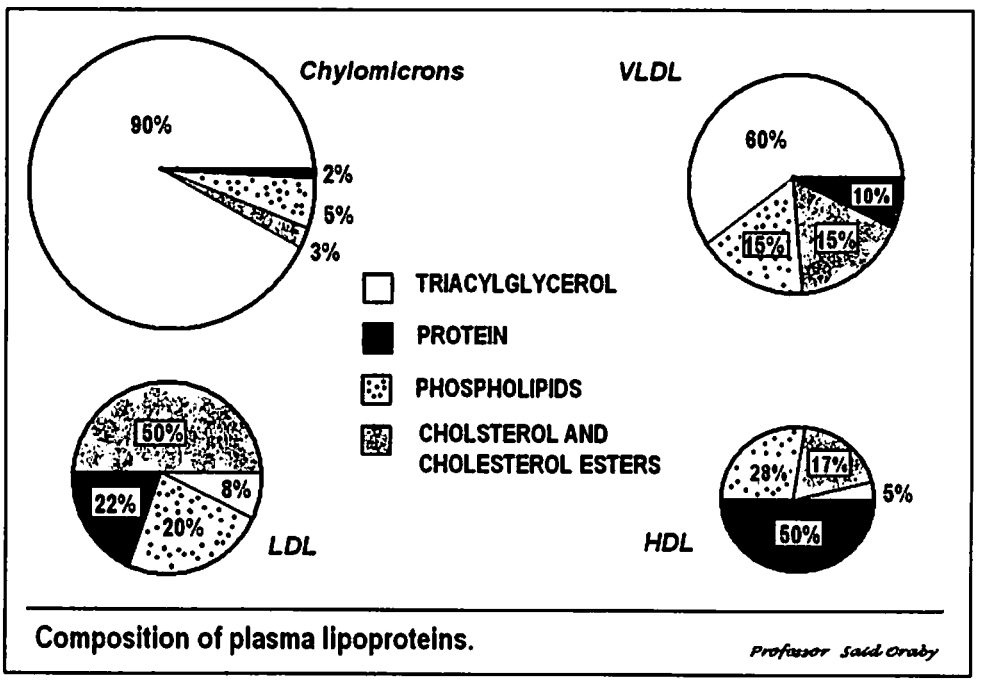
Seperation of plasma lipoproteins.

C. Fractions of plasma lipoproteins:

Each fraction of the five plasma lipoproteins (chylomicrons, VLDL, LDL, HDL and albumin free fatty acids complex) contains almost all types of lipids i.e. triacylglycerols, phospholipids, cholesterol, and free fatty acids. They differ in:

1. The main lipid content of each fraction.
2. Source of each fraction.
3. Types and amounts of associated proteins (apolipoproteins). The protein associated with fatty acids is albumin. The proteins associated with other types of lipids are globulins.

FRACTION	SOURCE	MAIN LIPID	APOLIPOPROTEINS	
			Amount	Types
Chylomicrons	Intestine	TG	2%	A, B_{48}, C &E.
VLDL	Liver	TG	10%	B_{100}, C & E.
LDL	Blood from chylomicrons and VLDL	Cholesterol, cholesteryl esters and phospholipids	22%	B_{100}
HDL	Liver	Cholesterol, cholesteryl esters and phospholipids	50%	A, C, D &E.
FFA-Albumin	Adipose tissue	FFA	99%	Albumin



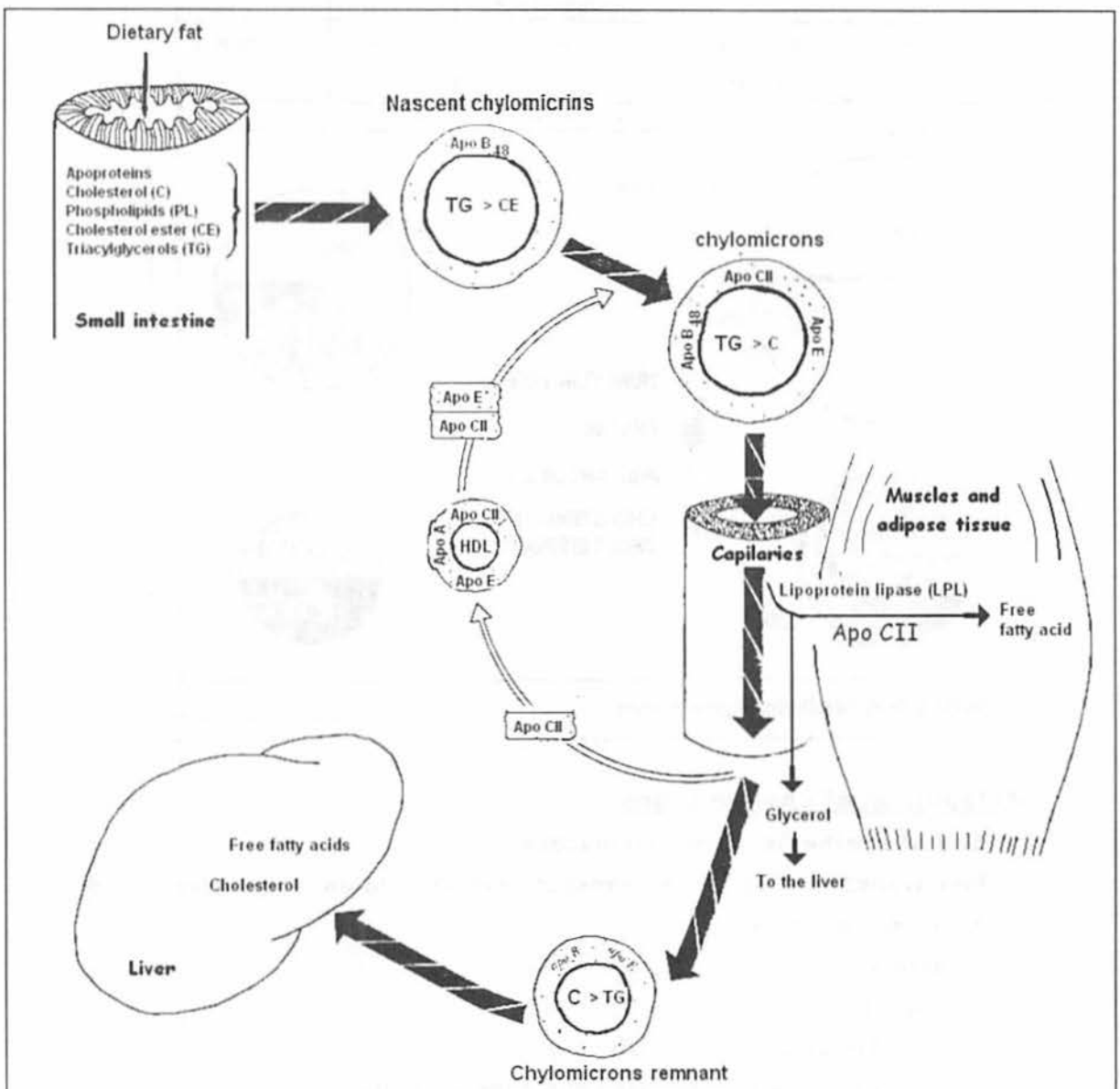
Composition of plasma lipoproteins.

Professor Said Orabi

D. Metabolism of chylomicrons:

1. **Site of synthesis:** Intestinal mucosa.
2. **Functions:** Chylomicrons transport **dietary lipids** from intestine ➔ Blood ➔ Peripheral tissues.
3. **Structure:**
   a) **Lipids:**
      1) Triacylglycerols, TG (main component).
      2) Cholesterol ester (CE) and phospholipids.
      3) Fat soluble vitamins.
   b) **Proteins:** 2% and include apo B_{48}, apo E and apo C (CII).

4. **Catabolism:**
   - *a)* The particles released by intestinal mucosal cells are called "nascent" chylomicrons and contains apolipoprotein B_{48}.
   - *b)* When it reaches the plasma, it receives -from circulating HDL- two apolipoproteins E and C and converted into chylomicrons.
   - *c)* The triacylglycerols component of chylomicrons will be hydrolyzed into glycerol and fatty acids. This reaction is catalyzed by lipoprotein lipase enzyme, which is activated by apo CII.
   - *d)* After hydrolysis of TG, the chylomicron particles begin to shrink. In addition, the apo CII returns to HDL. The remaining particles are called chylomicron remnants.



Metabolism of chylomicrons.

e) Chylomicron remnants are removed from circulation by the liver. Hepatocyte membranes contain receptors that recognize the both apo E and apo B_{48}.

f) The cholesterol released from chylomicron remnants in liver appears to regulate the rate of cholesterol synthesis by inhibiting HMG CoA reductase.

Summary of chylomicrons metabolism:
1-Site of synthesis: intestinal mucosal cells
2-Functions: transport dietary lipids from intestine to peripheral tissues.
3-Structure:
a-Main lipids: triacylglycerols. Chylomicrons contains also cholesterol, phospholipids and fat soluble vitamins.
b-Proteins: (2%), apo B_{48} and receives apo CII and apo E from HDL.
4-catabolism: TG are hydrolyzed by lipoprotein lipase (which is activated by apo CII). The remaining parts are chylomicron remnants, which are then taken up by the liver. Hepatocyte receptors can recognize apoB_{48} and apo E.

5. **Disorders of chylomicrons metabolism:**
   a) Deficiency of lipoprotein lipase: Leads to **hyperlipoproteinemia.**
   b) The disease is called **familial lipoprotein lipase deficiency**; it is characterized by marked increase of plasma chylomicrons, especially after fatty meal.

E. Metabolism of very low density lipoproteins (VLDL):

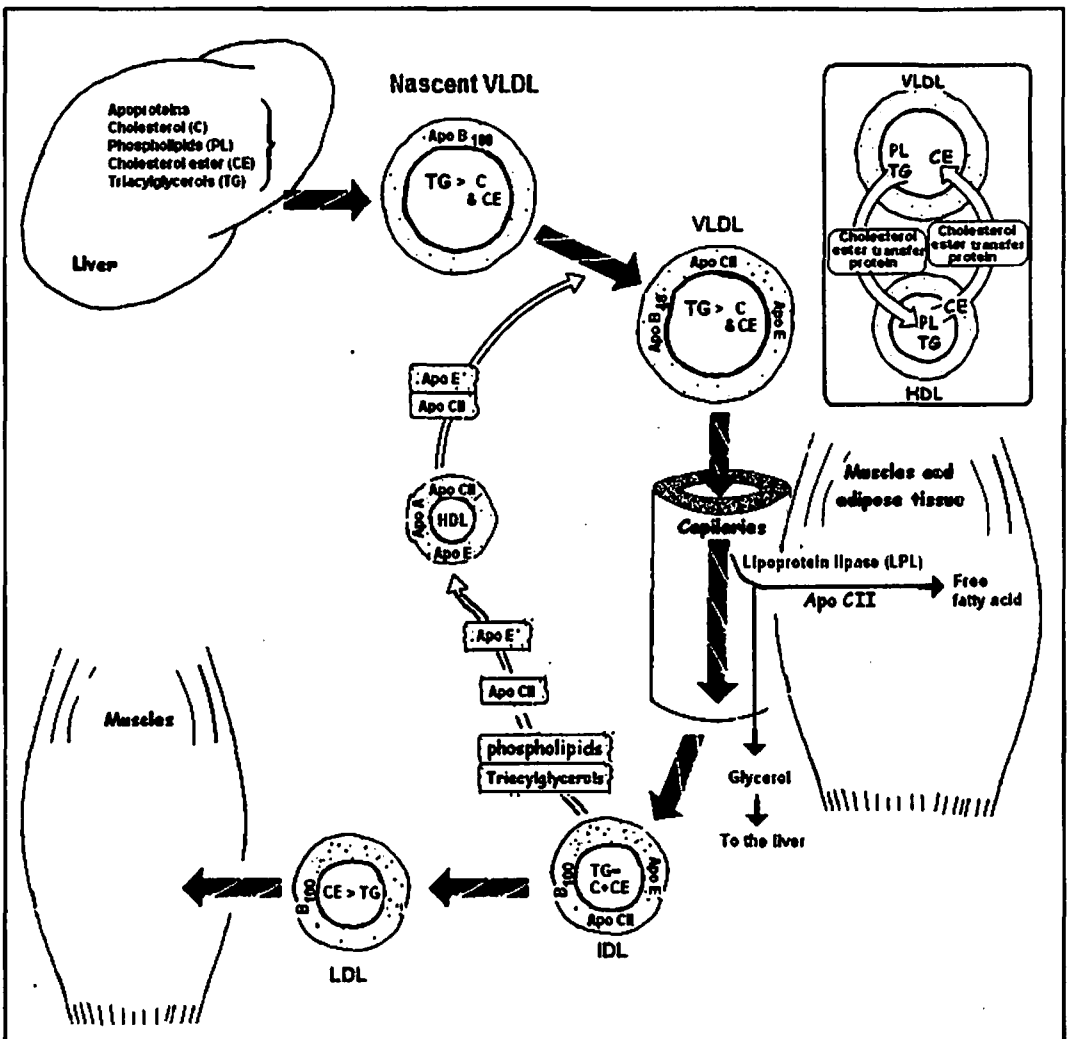
1. **Site of synthesis:** Liver
2. **Functions**: VLDL carries lipids from liver to the blood to the peripheral tissues.
3. **Structure:**
   a) **Lipids:**
      1) Triacylglycerols, TG (main component).
      2) Cholesterol esters and phospholipids.
   b) **Proteins:** 10% and include : apo B_{100}, apo E and apo C (CII)
4. **Catabolism:**
   a) The particles released from the liver are called **nascent VLDL** and contain B_{100}.
   b) When it reaches the plasma, it receives- from circulating HDL- two apolipoproteins; E and C and converted into VLDL.
   c) The triacylglycerols component of VLDL will be hydrolyzed into glycerol and fatty acids. This reaction is catalyzed by lipoprotein lipase enzyme, which is activated by apo CII.
   d) After hydrolysis of TG, the particles begin to shrink. The remaining particles are called **intermediate density lipoproteins (IDL).**

e) IDL is converted into LDL by transferring phospholipids, apo CII, and apo E to HDL.

f) LDL are removed from circulation by tissues as muscles (discussed later in metabolism of LDL).

g) VLDL particles also transfer phospholipids (PL) and TG to HDL. At the same time HDL transfer cholesterol esters (CE) to VLDL. These reactions are catalyzed by cholesterol ester transfer protein.

Summary of VLDL metabolism:
1-Site of synthesis: Liver.
2-Functions: transport lipids mainly TG from liver to peripheral tissues.
3-Structure:
a-Main lipids: triacylglycerols. It contains also cholesterol, phospholipids.
b-**Proteins:** (10%), apo B_{100} and receives apo CII and apo E from HDL.
4-catabolism: TG are hydrolyzed by lipoprotein lipase (that is activated by apo CII). The remaining parts are IDL, which are then converted into LDL by transferring phospholipids, apo CII and apo E to HDL.



Metabolism of VLDL.

5. **Disorders of VLDL metabolism:**
   a) **Fatty liver:** This is an accumulation of abnormal amounts of fat in liver. It occurs when there is excess triacylglycerols synthesis in the liver, which is then excreted in the form of VLDL.

F. Metabolism of low density lipoproteins (LDL):

1. **Site of synthesis**: circulation, from VLDL.
2. **Function**: The primary function of LDL particles is to provide cholesterol to the peripheral tissues. **They do so through**:
   a) Deposition of free cholesterol on cell membranes.
   b) By binding to receptors on cell membranes that recognize apo B_{100}.
3. **Structure:**
   a) Lipids: cholesterol, cholesteryl esters, and phospholipids.
   b) Apoproteins (22%): include apo B_{100}.
4. **Catabolism:**
   a) The numbers between brackets refer to the corresponding numbers on the following figure:

   [1] The LDL receptors are negatively charged glycoprotein molecules, made by the DNA of the cell. They are grouped in pits in cell membranes. The intracellular side of the pits is coated with a protein called **clathrin.**

   [2] After binding with receptors, the LDL are internalized as intact particles by endocytosis.

   [3] The vesicle containing the LDL rapidly loses its clathrin coat and fuses with other similar vesicles, forming large vesicle called **endosomes.**

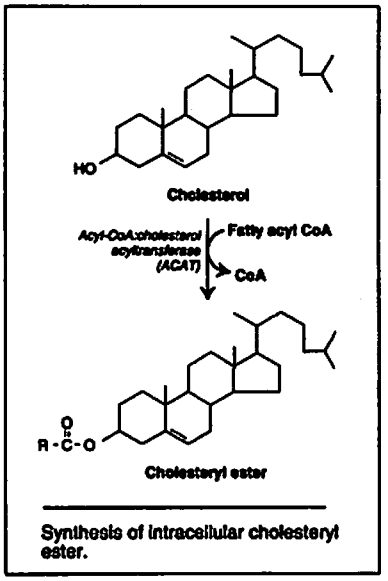
   [4] The pH of the contents of endosomes falls, allowing separation of the LDL from its receptors. The receptors then migrate to one side of the endosome while the LDL stays free within the lumen of the vesicle.

   [5] The receptors can be returned to the cell membrane. The lipoprotein remnants are hydrolyzed by lysosomal enzymes releasing cholesterol, amino acids, fatty acids and phospholipids.

   b) **If cell contains oversupply of cholesterol** : from LDL, HDL or chylomicron remnants, the cholesterol amount can be decreased and regulated by one of the following mechanisms:
      1) Oversupply of cholesterol inhibits HMG-CoA reductase activity so that cholesterol synthesis will be decreased.

2) Oversupply of cholesterol can stimulate a liver enzyme called: **acyl CoA cholesterol acyl transferase (ACAT):** ACAT transfers a fatty acid from fatty acyl CoA to cholesterol to form cholesteryl ester that can be stored inside the cell for future use.

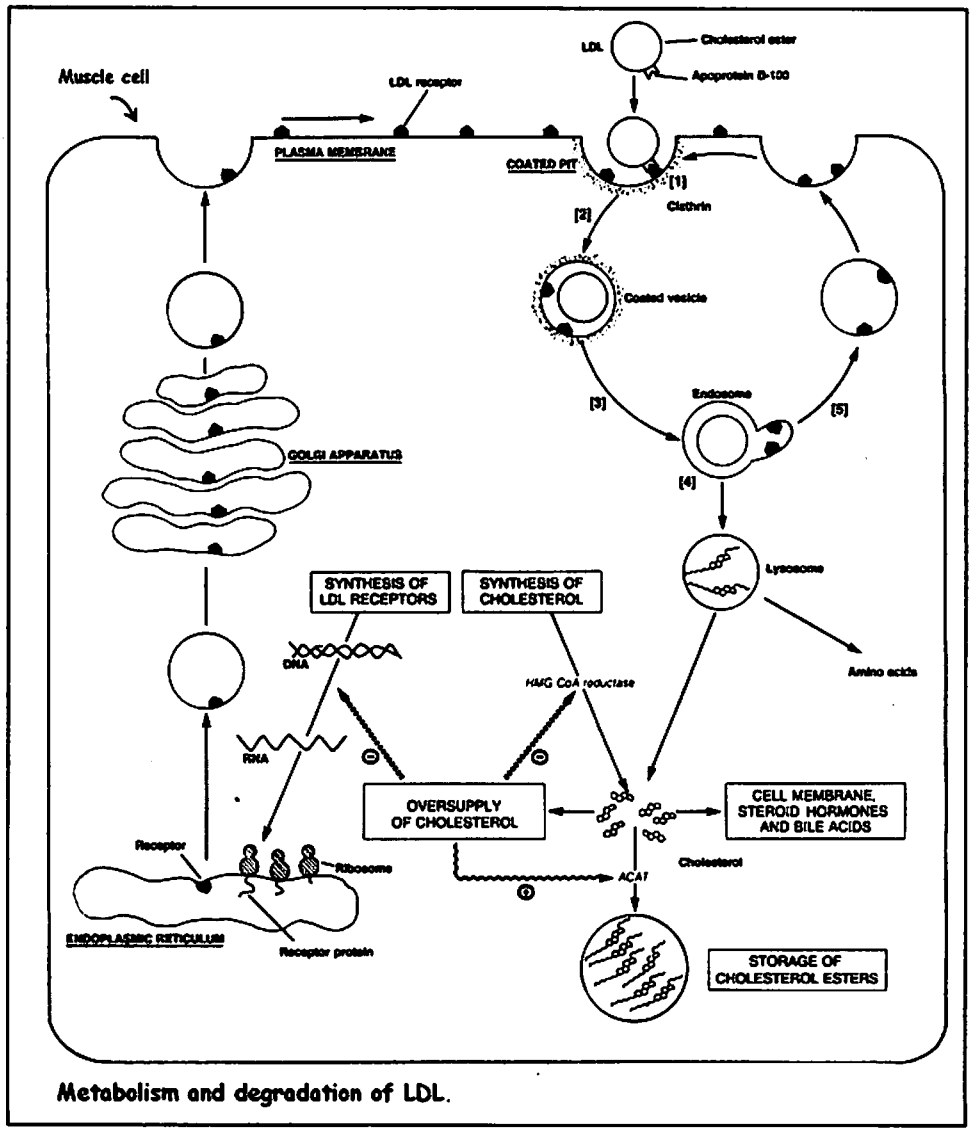
3) Oversupply of cholesterol inhibits the synthesis of new LDL receptor proteins, so that further entry of LDL cholesterol into the cell is limited.



Synthesis of intracellular cholesteryl ester.

5. Disorders of LDL metabolism:

a) **Type II familial hypercholesterolemia.:** LDL receptors are deficient in tissues and liver. Usually associted with atherosclerosis.

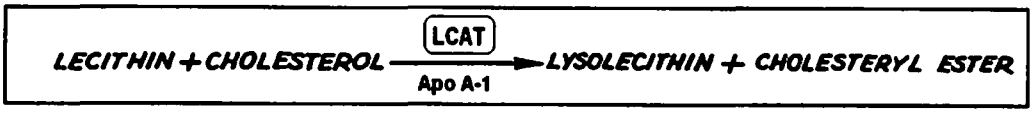


Metabolism and degradation of LDL.

SUMMARY OF LDL METABOLISM:
1-SITE OF SYNTHESIS: circulation from VLDL.
2-FUNCTION: LDL particles provide cholesterol to peripheral tissues.
3-STRUCTURE:
a- Lipid contents: cholesterol, cholesterol esters and phospholipids.
b- Protein contents: (22%), apo B_{100}.
4-CATABOLISM:
LDL apo B_{100} are recognized by tissue receptors. After binding with receptors, the LDL are internalized by endocytosis. Inside cells LDL are separated from receptors and hydrolyzed by lysosomal enzymes releasing cholesterol, amino acids, fatty acids and phospholipids.
***If the cell contains oversupply of cholesterol from LDL, HDL or chylomicron remnants, the cholesterol amount can be decreased by:**
a-Inhibition of HMG-CoA reductase ➔ Inhibition of cholesterol synthesis.
b-Stimulation of ACAT enzyme ➔ Cholesterol ester.
c-Inhibition of synthesis of LDL receptors ➔ inhibition of LDL uptake by cells.

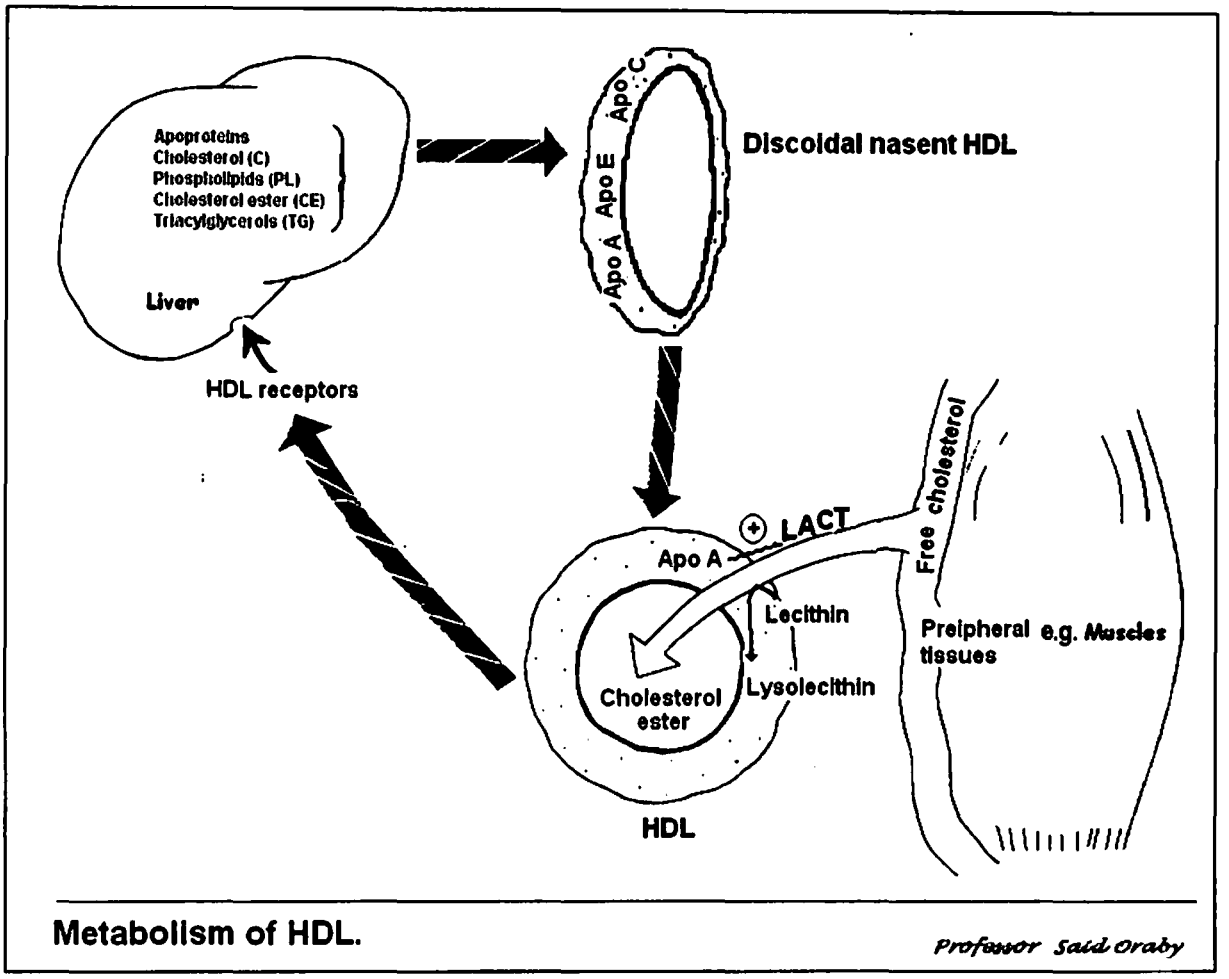
G. Metabolism of high density lipoproteins (HDL):

1. **Site of synthesis**: liver.
2. **Functions:**
   a) Act as reservoir of apo C-II that is transferred to chylomicrons and VLDL to activate lipoprotein lipase enzyme.
   b) Remove free (unesterified) cholesterol from extrahepatic tissue and esterifying it, using a plasma enzyme called: **lecithin, cholesterol acyl transferase (LCAT)**. The apo A-1 OF HDL activates LCAT.



   c) HDL particles carry cholesterol esters to:
      1) VLDL and LDL.
      2) Liver where the HDL is hydrolyzed and cholesterol released.
3. **Structure:**
   a) Lipids: Mainly phospholipids together with esterified & unesterified cholesterol.
   b) Proteins (50%): Include Apo A-1, Apo C and Apo E.
4. **Catabolism :**
   a) A newly secreted HDL are disc shaped particles containing mainly unesterified cholesterol and phospholipids.
   b) HDL is converted into spherical particles by accepting unesterified cholesterol from peripheral tissues (surface of cell membranes).
   c) Once the free cholesterol is taken up, it is immediately esterified by LCAT. The resulting cholesterol ester is very hydrophobic, so it remains in HDL and cannot be transferred to peripheral tissues.
   d) The liver takes up HDL particles, where the cholesteryl esters are hydrolyzed. The released cholesterol may undergo:

1) Binding with apoproteins to form lipoproteins.
2) Converting to bile acids.
3) Secreted into the bile to be removed from the body.



Metabolism of HDL.

Professor Said Oraby

Summary of HDL metabolism:

1-Site of HDL synthesis: Liver.

2-Functions:

a-Contain Apo C-II, which activates lipoprotein lipase that hydrolyzes TG.

b-Remove cholesterol from peripheral tissues and esterified it by LCAT enzyme ➔**Cholesterol esters.**

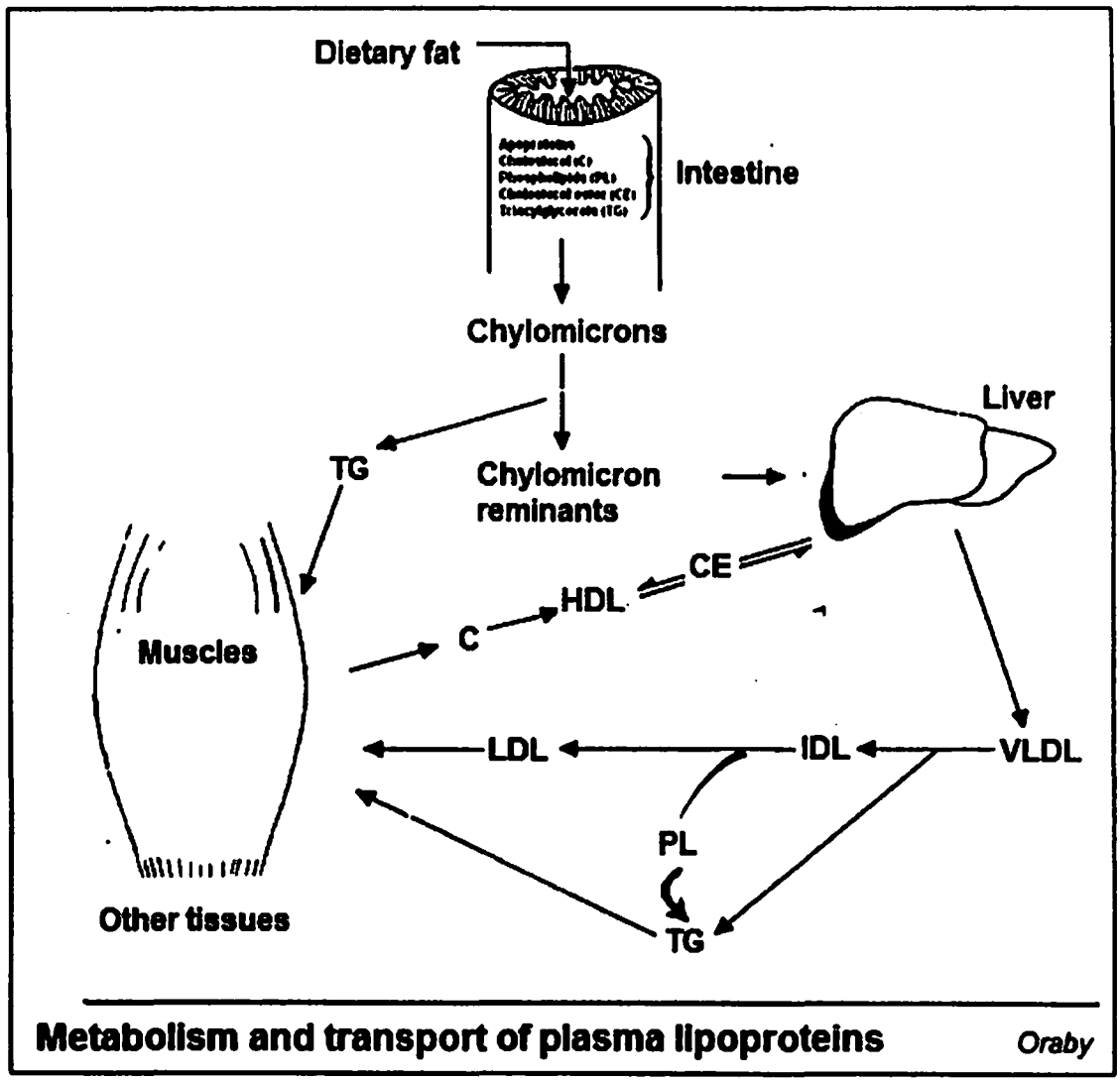
c- Carry cholesterol esters to VLDL & chylomicrons to the liver.

3-Structure:

a-Main lipids: cholesterol (free and esterified) and phospholipids, mainly lecithin.

b-**Proteins**: (50%), apo A-1(which activates LCAT), apo CII (which activates lipoprotein lipase and E (which is recognized by hepatic receptors)

4-Catabolism: HDL accepts unesterified cholesterol from peripheral tissues. Then HDL particles (by LCAT enzyme) esterify cholesterol into cholesterol esters. Then HDL is taken up by the liver cells where Cholesterol esters are released inside them to be utilized in the formation of lipoproteins or excreted in bile.



Metabolism and transport of plasma lipoproteins *Oraby*

H. Catabolism of free fatty acids (FFA):

1. Free fatty acids (FFA) are transported conjugated with albumin.
2. Plasma FFA are produced by lipolysis in adipose tissue and by the action of lipoprotein lipase in chylomicrons and VLDL.
3. Their level increases in all conditions associated with excessive lipolysis e.g. fasting, diabetes mellitus...etc.
4. Different tissues oxidize them to give energy.
5. In cases of excessive lipolysis e.g. uncontrolled diabetes mellitus, FFA are partially oxidized and produce ketone bodies.

I. *Lipoprotein (a):*

1. Lipoprotein A, commonly called Lp(a), is a major independent risk factor for cardiovascular disease. The optimum laboratory level should be under 20 mg/dl

2. Lipoprotein (a) is derived in the blood from low density lipoprotein (LDL) molecules and glycoprotein molecules called apolipoprotein-a (apo-a). Plasma apo-a is secreted by the liver.

3. High Lp (a) in blood is present in familial hypercholesterolemia, and is a risk factor for coronary heart disease (CHD), cerebrovascular disease (CVD), atherosclerosis, thrombosis, and stroke.

III. *APOPROTEINS (APOLIPOPROTEINS):*

A. **Definition:** These are proteins (globulins) present in association with plasma lipids to form lipoproteins.

B. **General role (functions):**

1. Apolipoproteins form with lipids water-soluble compounds, so they help transport of lipids between tissues.
2. Some apoproteins activate certain enzymes e.g. Apo C II activates lipoprotein lipase and Apo-A-1 activates LCAT.
3. They act as ligands (connection) for interaction of lipoprotein with their receptors in tissues i.e. receptors of lipoproteins in tissues can recognize lipoproteins through their apoproteins e.g. apo B_{100} for LDL receptors

C. **Classification (Types):**

Apolipoproteins	Lipoproteins	Functions
Apo A-1	HDL	• Activator of LCAT • Ligand for HDL receptor
Apo B_{-48}	Chylomicrons-Chylomicron remnants	• Synthesized by intestine • Ligand for chylomicron remnant receptors in liver
Apo B_{-100}	LDL – VLDL - IDL	• Synthesized by liver • Ligand for LDL receptors
Apo CII	Chylomicrons - VLDL - HDL	• Activator of lipoprotein lipase
Apo D	HDL	• May act as lipid transfer protein.
Apo E	Chylomicrons-Chylomicron remnants – VLDL - HDL	• Ligand for chylomicron remnant receptors in liver

***Note:** Apo B_{100} is synthesized in liver and is one of longest polypeptide chain (4530 amino acids), while Apo B_{48} is synthesized in intestine and is 48% as large as Apo B 100.*

IV. *DYSLIPOPROTEINEMIAS*: (DISORDERS OF PLASMA LIPOPROTEINS):

A. **Hyperlipoproteinemias:**

1. **Primary hyperlipoproteinemia:**

a) **Type I or familial lipoprotein lipase deficiency:** Chylomicrons and VLDL are markedly increased

b) **Type II or familial hypercholesterolemia:**

1) LDL receptors are deficient in tissues and liver
2) Usually associated with atherosclerosis.

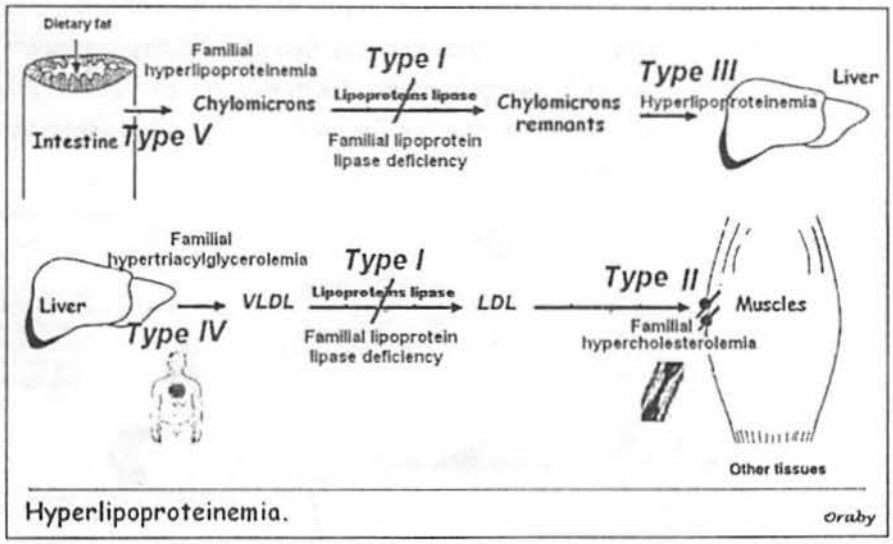
c) **Type III or hyperlipoproteinemia:**

1) Clearance of chylomicron remnants and VLDL remnants is deficient.
2) Chylomicrons and VLDL remnants are increased.

d) **Type IV or familial hypertriacylglycerolemia:**

1) Due to overproduction of VLDL.
2) Usually associated with coronary heart disease, type II diabetes and obesity.

e) **Type V or familial hyperlipoproteinemia:** Due to overproduction of VLDL and chylomicrons.



Hyperlipoproteinemia.

2. **Secondary hyperlipoproteinemia:** These abnormalities associated with other diseases as:

a) Diabetes mellitus
b) Hypothyroidism.
c) Nephrotic syndrome.
d) Obesity.
e) Obstructive jaundice.

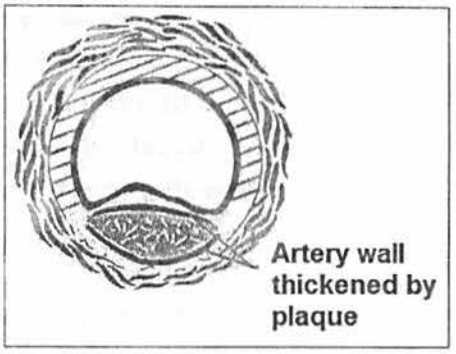
B. Hypolipoproteinemias:

1. **Abetalipoproteinemia:** Characterized by absence of LDL (β-lipoprotein). It is associated with low concentrations of chylomicrons and VLDL.
2. **Tangier disease:**

a) It is due to deficiency of **LCAT** enzyme.
b) It is characterized by low concentration of HDL with accumulation of cholesterol in tissues.

V. *Plasma lipoproteins and atherosclerosis:*

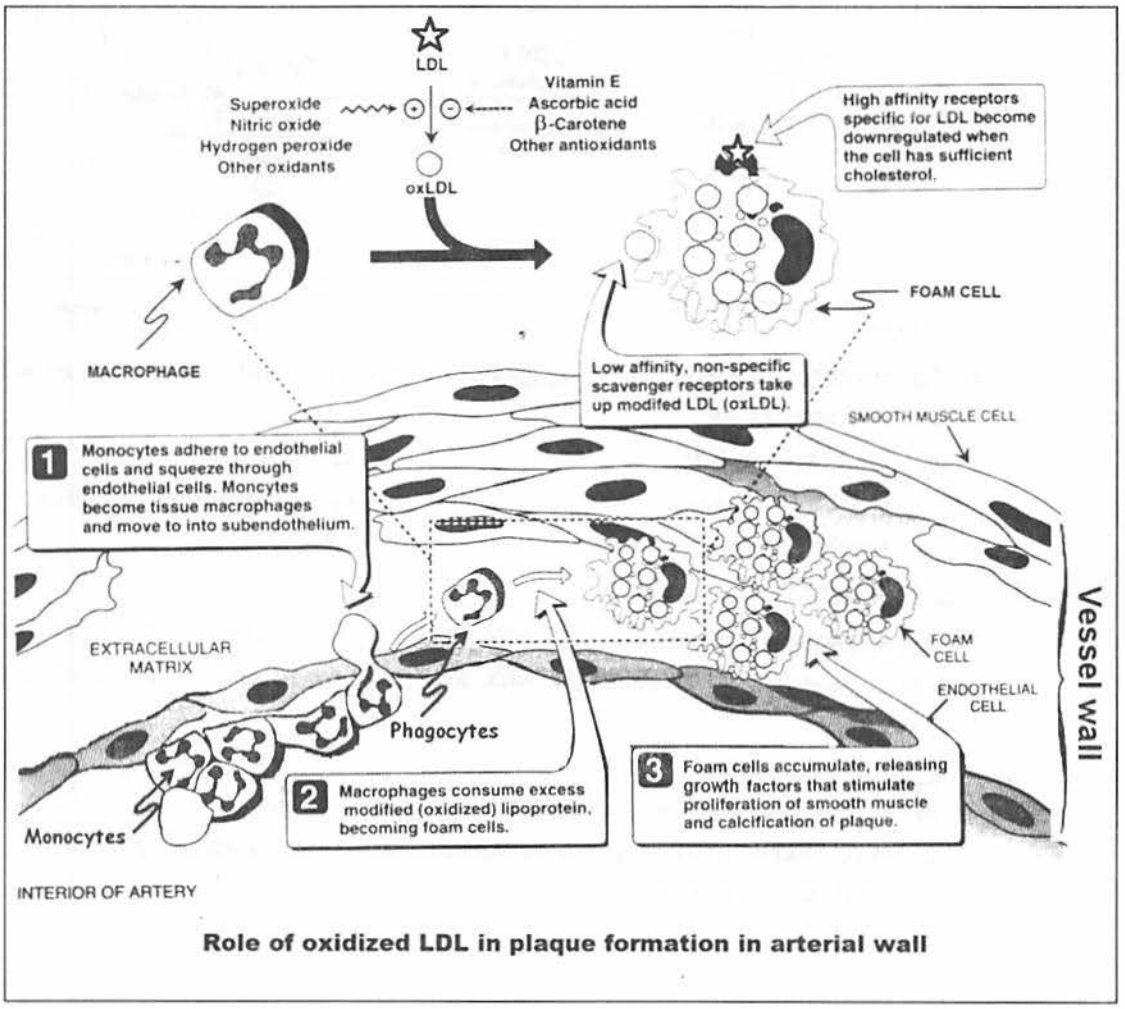
A. Definition: Atherosclerosis is a chronic disease in which deposits of cholesterol, cholesteryl esters and cellular debris accumulate in the inner surfaces of large and medium sized arteries forming plaque. As the disease progresses, the deposits reduce or even, stop the flow of blood causing **coronary artery disease** or **cerebral artery disease**.



NOTE: If atherosclerosis affects and blocks coronary artery branches of the heart, it may lead to death of heart cells and this is called myocardial infarction.

B. Causes of atherosclerosis:

1. Diseases associated with prolonged elevated levels of VLDL, IDL and LDL e.g. diabetes mellitus; hypothyroidism and hyperlipidemia.
2. Oxidation of LDL by superoxide, hydrogen peroxide and other oxidants into oxidized LDL ➔ Plaque formation and atherosclerosis. See the following diagram:



Role of oxidized LDL in plaque formation in arterial wall

C. **Risk factors for atherosclerosis:** Since the LDL represents the transport of cholesterol to the tissues and HDL represents the removal of cholesterol from tissues thus:

1. **LDL/HDL ratio** helps in predicting atherosclerosis and coronary heart disease:

   ↑ LDL/HDL → Atherosclerosis

   ↓ LDL/HDL → No atherosclerosis

2. ***Estrogens and exercises cause increase HDL***, thus premenopausal women and persons doing exercises (jogging) appear to be protected from coronary heart diseases. Also, diet rich in polyunsaturated fatty acids lowers blood cholesterol.

Summary of lipase enzymes:

Enzyme	Origin	Site of action	Function	Special properties
Gastric lipase	Stomach	Stomach	Degrades dietary triacylglycerols in infants	Needs acid pH
Pancreatic lipase	Pancreas	Small intestine	Degrades dietary triacylglycerols (removes fatty acid from carbon 1 and 3, leaving 2-monoglycerol)	Needs colipase
Lipoprotein lipase	Blood and extrahepatic tissues	Surface endothelial cells lining the capillaries	Degradeds triacylglycerol circulating in chylomicrons or VLDL, releasing FA and glycerol	Can be released into plasma by heparin; activated by apoprotein C-II
Hormone sensitive lipase	Adipose tissue	Adipose tissue	Degradation of stored triacylglycerols	Activated by cAMP
Acid lipase	Most tissues	Lysosomes	Removes fatty acids from lipids taken into cells during phagocytosis	Needs acid pH
Hepatic lipase	Liver	Liver	Removes fatty acids from chylomicron remnants and HDLtaken by hepatic cells.	

VI. *Role of liver in fat metabolism:*

1. Fatty acid synthesis and oxidation.
2. Synthesis and esterification of cholesterol.
3. Formation of lipoproteins.
4. Formation of phospholipids.
5. Formation of ketone bodies.
6. Formation of bile salts.
7. Storage of fat soluble vitamins.
8. Detoxication of steroid hormones.